HMI-01 LABVIEW SCADA FUNCTIONAL ARCHITECTURE

SUPERVISORY MONITORING AND PARAMETER CONTROL

SEE DWG 003 / 011 / 012 / 013 / 014

A. HMI SCREEN NAVIGATION ARCHITECTURE

AUTO OPERATION SCREEN
AUTO START / NORMAL STOP
CYCLE STATE / PROCESS STATUS
INFEED / TEST / SORT / OUTFEED STATUS

READ ONLY PLC STATE DISPLAY
S10 Ready / Idle
S20 Wait Infeed
S40~S60 Clamp / Test
S30 Transfer to Test
S70~S100 Release / Gate / Outfeed
SF0 Fault Hold

TREND / PRODUCTION SCREEN
CYCLE COUNT / PASS / FAIL / RETEST
CYCLE TIME / ALARM COUNT
TREND / CSV / TDMS LOG

PRODUCTION COUNT / CYCLE TREND

HMI-01
MAIN OVERVIEW SCREEN
SYSTEM STATUS / MODE / CURRENT ALARM

PLC COMMUNICATION STATUS
SAFETY STATUS
AUTO / MANUAL MODE
CURRENT STATE S00~S110
ACTIVE ALARM
TOTAL / PASS / FAIL / RETEST COUNT

MANUAL / SETUP SCREEN
MODE SELECT
SETPOINT ENTRY
COMMISSIONING COMMANDS

MANUAL COMMAND SHALL BE INHIBITED
WHEN SAFETY IS NOT READY OR SF0 FAULT IS ACTIVE.

ALARM / HISTORY SCREEN
ACTIVE ALARM
ALARM CODE / TIME / RESET STATUS
ALARM HISTORY / EVENT LOG

ACTIVE ALARM / ALARM HISTORY

SETTINGS / COMMUNICATION SCREEN
PLC LINK STATUS
MCU HEARTBEAT / MODBUS STATUS
HMI COMMUNICATION PARAMETER

READ ONLY DIAGNOSTIC DATA

B. LABVIEW HMI / PLC-01 COMMUNICATION FLOW

HMI-01
MAIN OVERVIEW SCREEN
SYSTEM STATUS / MODE / CURRENT ALARM
SCREEN / ALARM / TREND / SETPOINT / HISTORY

SETPOINT / MODE / HMI COMMAND

RS-232C
HMI READ / WRITE DATA LINK

STATE / I-O / ALARM / COUNT / COMMUNICATION STATUS

PLC-01
HMI DATA INTERFACE
RS-232C / XGT PROTOCOL

ACTIVE ALARM / HISTORY DISPLAY

C. HMI DATA OWNERSHIP AND COMMAND FLOW

READ ONLY STATUS

STATE / I-O / FAULT / MCU STATUS

OPERATOR COMMAND REQUEST

OPERATOR SETPOINT WRITE

PLC VALIDATED SETPOINT

PLC REQUEST PROCESSING

CYCLE / PASS / FAIL / RETEST COUNT

ALARM / FAULT / RESET STATUS

PLC TO HMI STATUS DATA
M0100~M0114 STATE STATUS
M0120 FAULT HOLD
P00~P0B INPUT STATUS
P20~P29 OUTPUT STATUS
D0020~D0023 TEST / MCU STATUS

HMI TAG, COMMAND, DATA LOGGING,
AND GENERAL HMI NOTES:
SEE DWG 015 / SHEET 2 OF 3.

HMI TO PLC SETPOINT DATA
D0050 TEST / POSITION SETPOINT
D0051 MOTOR SPEED SETPOINT
D0052 PROCESS / VIBRATION THRESHOLD
0 = AUTO / 1 = MANUAL
D0053 MODE SELECT

HMI COMMAND BITS SHALL BE PROCESSED
AS PLC REQUESTS ONLY.
SEE DWG 006 / 013.

HMI TO PLC COMMAND DATA
START REQUEST
NORMAL STOP REQUEST
ALARM RESET REQUEST
COMMAND PERMITTED ONLY
WHEN SAFETY READY

HMI COMMAND SHALL NOT REPLACE
HARDWIRED E-STOP OR
GUARD DOOR SAFETY FUNCTION.

PLC TO HMI ALARM / HISTORY DATA
ALM-001 / ALM-101~ALM-108
ALARM ACTIVE / RESET STATUS
CYCLE / PASS / FAIL / RETEST COUNT
TIME STAMP / EVENT LOG

ALARM EVENT / CYCLE DATA / TREND LOG
LOGGED BY HMI-01

DATA LOGGER
CSV / TDMS / TREND FILE

NO.	DATE	REVISION
A	2026-09-16	INITIAL RELEASE

DWG TITLE

EDS Wafer Map Linked Package
Final Test and Multi-Bin Sorting Cell

SCADA HMI FUNCTIONAL ARCHITECTURE

DRAWN BY	DATE
J. H. KO	2026-09-16

DWG NO
015

SHEET NO
1 OF 3